LAB ASSIGNMENT 9

Mapping C Code to Digital Circuits Using Generative AI

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1. GenAI Model Name

ChatGPT (GPT-6)

2. Prompt

Use Generative AI to explain how a small C program can be represented using digital logic circuits. Identify the required logic gates, registers, arithmetic circuits, and control signals, and explain their functions in simple language.

3. Response

Consider the following C program:

```
int a = 3;

int b = 2;

int c;

c = a + b;
```

This program adds two numbers and stores the result in variable c.

Digital Circuit Mapping:

- **Input Registers:** Store the values of a and b.
- **4-bit Adder:** Adds the two input values.
- **Output Register:** Stores the result in c.
- **Clock Signal:** Controls when the output register stores the result.
- **Control Signal:** Enables the circuit to perform the required operation.

For the given program:

A = 0011
B = 0010

The adder performs:

0011 + 0010 = 0101

Therefore, the result stored in c is 0101, which represents decimal 5.

4. Verification by Logisim

The circuit can be implemented in Logisim Evolution using two 4-bit input pins, one 4-bit adder, and one 4-bit output pin.

Test Case:

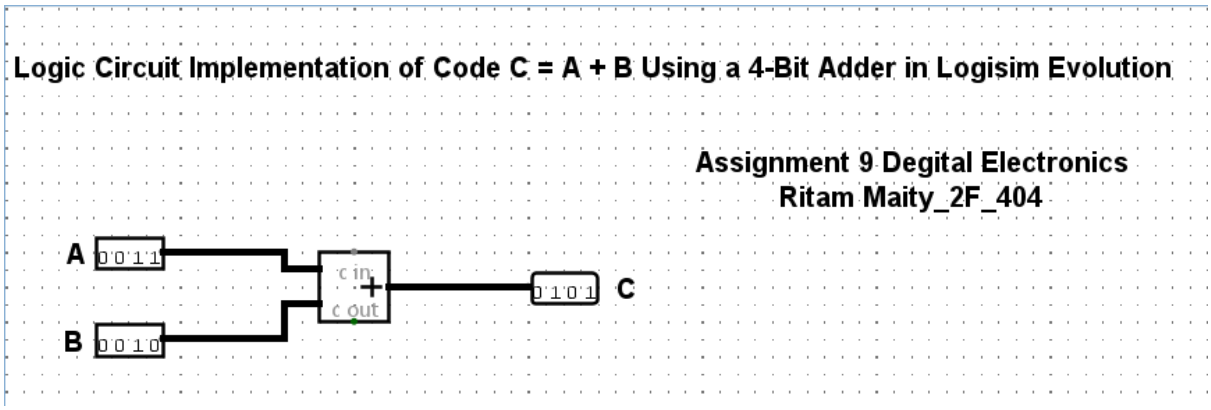
Input A	Input B	Expected Output C
0011	0010	0101
0100	0001	0101
0010	0010	0100

The circuit is considered correct when the simulated outputs match the expected results.

5. Circuit Diagram

Figure 1: C Code to Digital Circuit Mapping

Connect the 4-bit input A and 4-bit input B to the two inputs of a 4-bit adder. Connect the adder's Sum output to a 4-bit output pin labelled C.



6. Output

For input A = 0011 and input B = 0010, the adder produces output C = 0101. This represents the result of the C statement c = a + b;.

7. Discussion

This assignment demonstrates how a simple C arithmetic statement can be represented using digital hardware. The addition operation is implemented using an adder, while registers can store the input values and the result. Generative AI helps explain the relationship between software instructions and digital circuit components.

8. GitHub Link

GitHub Repository: https://github.com/Ritam-Maity/Digital-Electronics-Lab